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SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Symbiosis International (Deemed University)

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Assignment No. 13

Subject:	DevOps Lab
TCODE:	TE7950
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Branch	CSE, Batch (2023-27), L4
Academic Year & Semester	25 – 26, Sem VII
Date of Performance	23rd July, 2026
Title of Assignment:	Kubernetes Service & Networking
Theory:	A Kubernetes Service is an abstraction that provides a stable network endpoint for accessing one or more Pods in a Kubernetes cluster. Since Pods are temporary and their IP addresses may change, a Service uses labels and selectors to route incoming traffic to the appropriate Pods. A NodePort Service exposes the application outside the cluster by allocating a port on each cluster node, allowing external access to the application. During local development, kubectl port-forward can also be used to securely forward traffic from a local machine to a Kubernetes Service for testing and verification. This approach simplifies communication between applications and ensures reliable access to containerized workloads. **Commands Used** • Created the Kubernetes Deployment: kubectl apply -f web-deployment.yaml • Verified that the Pods were running: kubectl get pods • Created the Kubernetes Service: kubectl apply -f web-service.yaml • Displayed the available Kubernetes Services: kubectl get service • Displayed detailed information about the Service: kubectl describe service web-app-service • Displayed the Service endpoints: kubectl get endpoints

	• Displayed the running Pods with labels: kubectl get pods --show-labels • Displayed the application logs: kubectl logs web-app-deployment-<pod-name> • Forwarded the Kubernetes Service to the local machine: kubectl port-forward service/web-app-service 8085:80 • Deleted the Kubernetes Service: kubectl delete -f web-service.yaml • Deleted the Kubernetes Deployment: kubectl delete -f web-deployment.yaml
Implementation:	**Step 1: Create the Kubernetes Service YAML** Created Kubernetes Service Manifest — Created the web-service.yaml file defining a Kubernetes NodePort Service to expose the Flask web application running in the Kubernetes Pods. **Step 2: Apply the Servic** Applied Kubernetes Service — Applied the Kubernetes Service manifest using kubectl apply, creating a NodePort Service for the deployed Flask application. **Step 3: Find the Service Information** Verified Kubernetes Service — Verified the Kubernetes NodePort Service and identified the assigned NodePort (30649) used to expose the Flask web application outside the Kubernetes cluster. **Step 4: Access the Application** Accessed Application Through Kubernetes Service — Used kubectl port-forward to forward the Kubernetes Service to the local machine and successfully accessed the Flask web application at http://localhost:8085, verifying that it was running correctly. **Step 5: Clean Up** Deleted Kubernetes Service — Removed the Kubernetes Service using the kubectl delete command, cleaning up the networking resources created for the application.

docker-compose.yml U | ! deployment.yaml U | ! service.yaml U

! service.yaml

```yaml
apiVersion: v1
kind: Service

metadata:
  name: flask-service

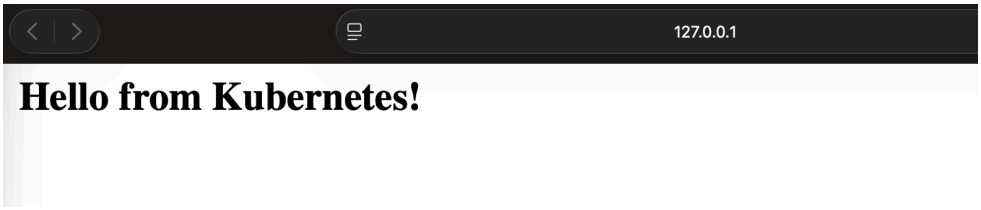
spec:
  selector:
    app: flask

  type: NodePort

  ports:
    - protocol: TCP
      port: 5000
      targetPort: 5000
      nodePort: 30080
```

```
yashmitkotekar@Yashmits-MacBook-Air Task2-Kubernetes % kubectl apply -f deployment.yaml
deployment.apps/flask-deployment unchanged
yashmitkotekar@Yashmits-MacBook-Air Task2-Kubernetes % kubectl get deployments
NAME               READY   UP-TO-DATE   AVAILABLE   AGE
flask-deployment   1/1     1            1           33m
yashmitkotekar@Yashmits-MacBook-Air Task2-Kubernetes % kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
flask-deployment-59c5fd8d77-wrrts   1/1     Running   0          4m25s
yashmitkotekar@Yashmits-MacBook-Air Task2-Kubernetes %
```

```
yashmitkotekar@Yashmits-MacBook-Air Task2-Kubernetes % kubectl rollout restart deployment flask-deployment
deployment.apps/flask-deployment restarted
yashmitkotekar@Yashmits-MacBook-Air Task2-Kubernetes % kubectl get pods
NAME                                READY   STATUS        RESTARTS   AGE
flask-deployment-59c5fd8d77-wrrts   1/1     Running       0          15s
flask-deployment-86b795544f-flg2c   1/1     Terminating   0          29m
yashmitkotekar@Yashmits-MacBook-Air Task2-Kubernetes % minikube service flask-service
|-----------|---------------|-------------|---------------------------|
| NAMESPACE |     NAME      | TARGET PORT |            URL            |
|-----------|---------------|-------------|---------------------------|
| default   | flask-service | 5000        | http://192.168.49.2:30080 |
|-----------|---------------|-------------|---------------------------|
🔗  Starting tunnel for service flask-service.
|-----------|---------------|-------------|------------------------|
| NAMESPACE |     NAME      | TARGET PORT |          URL           |
|-----------|---------------|-------------|------------------------|
| default   | flask-service |             | http://127.0.0.1:51724 |
|-----------|---------------|-------------|------------------------|
🎉  Opening service default/flask-service in default browser...
❗  Because you are using a Docker driver on darwin, the terminal needs to be open to run it.
```



Subject Faculty Name: Prof. Arijit Dutta